MILITARY SPECIFICATION

VOLTMETERS, ELECTRONIC

1. SCOPE

* 1.1 Scope. This specification covers portable, solid state, alternating current (AC) voltmeters.

* 1.2 Classification. Voltmeters covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Cabinet model.

Type II - Rack model.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-17555	Electronic And Electrical Equipment And Associated Repair Parts; Preparation For Delivery Of

STANDARDS

Military

MIL-STD-130	Identification Marking Of US Military Property

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Qualification. Instruments furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.9 and 6.3).

* 3.1.1 Sample articles furnished for qualification testing shall be manufactured by the prospective supplier requesting qualification and shall be constructed of the same parts, components, subassemblies, materials, and by the same manufacturing processes intended to be used during production. Production items

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shall exhibit the same quality and component reliability used in sample articles. Sample articles shall have been tested in accordance with the qualification provisions in 4.9, prior to submission for qualification testing.

3.1.2 Design changes occurring after qualification approval will require resubmission for qualification testing, unless a specific waiver is granted by the activity responsible for the qualified products list.

3.1.3 The requirements of this specification reflect the minimum characteristics and accuracy. Items that provide additional characteristics, greater accuracy, and improved performance will be acceptable.

3.2 Design and construction.

3.2.1 Design. All circuitry shall exhibit sufficient stability of operating parameters to maintain calibration within the allowable tolerance, without resetting any internal controls, for the environmental and reliability conditions as specified herein. The circuitry of this equipment shall be designed to preclude the necessity of sorting or selecting of any component part from within the stated tolerance of such part, during construction, testing, or subsequent maintenance actions.

3.2.2 Construction. The instrument shall be of modular miniaturized, solid state construction. The construction shall allow maximum accessibility to internal parts and subassemblies.

3.2.3 Adjustments. Adjustments or compensating devices, capable of restoring the circuitry to the specified tolerances, shall be provided for any function or parameter in which the combined time of shelf life and usage, for a period of 6 months, causes the instrument to exceed the specification tolerances.

3.2.4 Reliability. Instruments furnished under this specification shall have a specified mean time between failure (MTBF) of 1000 hours and shall be capable of passing the qualification reliability tests specified in 4.8. A failure as used herein is any need to recalibrate by any adjustment not considered an operating control in Section 4 of this specification, or any replacement of components, after the test is initiated, in order for the instrument to meet the requirements of this specification.

3.2.5 Acceptance. Instruments furnished under this specification shall be capable of meeting the quality conformance tests specified in 4.4 and the qualification reliability tests specified in 4.8.

3.3 Functional description. The voltmeters, hereafter called instruments, shall be capable of measuring AC voltages from 1 millivolt (mv) to 300 volts full scale, and decibels (dB) from minus 60 dB to plus 50 dB. The frequency range of the instrument shall be from 10 Hertz (Hz) to at least 4 megahertz (MHz). Unless otherwise specified, the overall accuracy of the instrument shall be within ±4 percent.

3.4 Composition. The instrument shall consist of the following items and other component parts specified herein and as required to make up a complete equipment:

a. Power supply (3.5.1).

b. AC voltmeter section (3.5.2).

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c. Amplifier (3.5.3).

d. Visual display (3.5.4).

e. Range selector switch (3.5.5).

f. Terminals (3.5.6).

g. Test leads (3.5.7).

3.5 Performance requirements.

3.5.1 Power supply. The power supply shall be capable of regulating the supply voltages for all power sources specified in 3.6.

3.5.2 AC voltmeter section.

3.5.2.1 Range. The AC voltmeter section shall provide a minimum of 12 ranges for measuring AC voltages from 1 mv to 300 volts full scale, and dB from minus 60 dB to plus 50 dB.

* 3.5.2.2 Frequency response. The frequency response of the instrument shall be from 10 Hz to at least 4 MHz.

* 3.5.2.3 Accuracy. The calibration accuracy of the instrument shall be as follows:

a. 10 Hz to 20 Hz, within ±4 percent of full scale.

b. 20 Hz to 50 Hz, within ±2 percent of full scale.

c. 50 Hz to 1 MHz, within ±1 percent of full scale.

d. 1 MHz to 4 MHz, within ±2 percent of full scale.

* 3.5.2.4 AC meter tracking. The AC meter tracking accuracy shall be within ± 2 percent of full scale.

3.5.2.5 Input impedance. The input impedance shall be at least 10 megohms, paralleled by not more than 25 picofarads (pf).

3.5.3 Amplifier.

* 3.5.3.1 Output voltage and impedance. The amplifier output circuit shall have an open circuit voltage of at least 150 mv root mean square (rms) at full scale deflection, with an internal impedance of not more than 60 ohms, and a direct current (DC) output of at least 1 volt on any range at full scale deflection.

3.5.3.2 Frequency response. The amplifier frequency response shall be as specified in 3.5.2.2.

3.5.3.3 Noise. The amplifier noise, with the input terminal(s) shorted together, shall be less than 3 percent of full scale on all ranges.

3.5.3.4 AC hum pickup. Hum pickup at 60 Hz, and harmonics thereof, shall be less than ±1 percent of full scale.

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3.5.4 Visual display,

3.5.4.1 Meter. The meter shall be calibrated in rms volts and shall indicate the rms value of an applied sine wave voltage. The meter shall have an edge mounted pointer and a mirror-backed scale with a minimum length of 5 inches. A device shall be provided for mechanically adjusting the meter pointer.

3.5.4.2 Scales. The meter shall have 2 voltage scales and 1 dB scale; the voltage scales will be the uppermost scales on the meter face. The arrangement of the voltage and dB scales shall be such that the dB increments, specified below, correspond to the adjacent voltage scale values, specified below, within ± 1 percent of full scale.

DB Scale	Voltage Scale
-12 dB	0.195
-11 dB	0.218
-10 dB	0.245
- 9 dB	0.275
- 8 dB	0.308
- 7 dB	0.347
- 6 dB	0.388
- 5 dB	0.436
- 4 dB	0.489
- 3 dB	0.548
- 2 dB	0.615
- 1 dB	0.690
0 dB	0.775
+ 1 dB	0.869
+ 2 dB	0.975

3.5.4.3 Voltage scales. The 2 voltage scales shall be 0 to 1 and 0 to 3. The zero point of both scales shall be at the same point on the meter face.

3.5.4.3.1 Markings. The 0 to 1 voltage scale shall be divided into not less than 100 scale divisions and shall have numerical markings at the following divisions: .2, .4, .6, .8, and 1. The 0 to 3 voltage scale shall be divided into not less than 60 scale divisions and shall have numerical markings at the following divisions: 1, 2, and 3.

3.5.4.4 DB scale. The meter shall have a dB scale calibrated from minus 12 to plus 2. The scale shall be arranged to indicate dB in reference in 1 milliwatt (dbm) across a 600 ohm load.

3.5.4.4.1 Markings. The dB scale shall be marked in 1 dB increments and shall have numerical markings at the following divisions: -10, -5, 0, and +2. Additional numerical markings may be used.

3.5.4.5 Switching transients. With the input terminal(s) shorted together, the meter deflection due to switching transients shall be less than 25 percent of full scale. The recovery time shall be less than 10 seconds.

3.5.4.6 Meter response. The response time of the meter pointer to reach full scale deflection from zero shall be not greater than 1.5 seconds.

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3.5.5 Range selector switch. The range selector switch shall be located on the front panel of the instrument and shall have at least 12 positions. Each position shall be numerically marked in volts and dB. The voltage markings shall be from 1 mv to 300 volts in a 1-3-10 sequence. The dB markings shall be from minus 60 dB to plus 50 dB in 10 dB steps. The switch shall have positive detent action and shall be aligned properly with the panel markings.

* 3.5.6 Terminals. An input terminal shall be provided on the front panel of the instrument. Output terminals, both AC and DC, shall be provided on the front or rear panel of the instrument. Terminals which accept shielded double banana type connectors shall have the "screw down" type binding posts or lugs.

* 3.5.7 Test leads. The instruments shall be provided with the necessary test lead(s) to perform the measurements as specified herein. The length of the test lead(s) shall be at least 40 inches, external to the instrument.

3.6 Electrical requirements.

* 3.6.1 Warm up. The instrument's warm up time, at temperatures down to 0° Centigrade (C), shall not exceed 2 minutes.

* 3.6.2 Power. The instrument shall be capable of operating within specifications with an input rms voltage of 115 and 230 volts, ±10 percent. The instrument shall accept and operate satisfactorily on any line voltage frequency of 50, 60, and 400 Hz, ±5 percent.

* 3.6.2.1 Power selection. A switching device, capable of selecting the desired input line voltage of 115 or 230 volts, shall be accessible from the front or rear panel of the instrument. The device shall have a positive locking action to prevent accidental selection of the improper line voltage. The 115 and 230 volt positions shall be clearly marked. The instrument shall be delivered with the power selection device in the 115 volt position and a proper fuse(s) installed. If a different size fuse is required for 230 volt operation, the proper fuse(s) shall be attached to the caution tag. The following caution tag shall be attached to the instrument:

"CAUTION: THIS INSTRUMENT IS CONNECTED FOR 115 VOLT OPERATION"

* 3.6.3 Power disconnect. A switching device and fuse(s) shall be installed in series with the power input to the instrument. The fuse(s) shall be externally accessible.

* 3.6.4 Power cable. A power input cable, either fixed or detachable, shall be provided. The power cable shall be fitted at one end with the Electronic Industries Association (EIA) recommended grounding type parallel blade, 3 wire male connector. Adapters shall be provided to convert from the grounding 3 wire connector to the nongrounding connector for 115 volt operation. The length of the power cable with connector shall be at least 68 inches, external to the instrument.

3.7 Mechanical requirements.

* 3.7.1 Dimensions.

* 3.7.1.1 Type I dimensions. Type I instruments procured under this specification shall be provided with a cabinet for bench use. The girth (face perimeter) plus the depth of the instrument shall be not greater than 40 inches.

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3.7.1.2 Type II dimensions. Type II instruments procured under this specification shall be capable of mounting in a 19 inch EIA rack and the panel height shall be a multiple of 1 3/4 inches, EIA standard panel size. The maximum height of the rack mount instrument shall be not greater than 8 3/4 inches. The maximum depth shall be not greater than 21 inches.

3.7.2 Weight. The unpacked weight of the instrument shall be not greater than 10 pounds.

3.7.2.1 Portability. The instrument shall be fitted with a carrying handle(s), capable of supporting the weight as specified in 3.7.2. The location of the handle(s) shall be such that the above weight is evenly distributed for easy handling.

3.7.3 Cooling. The instrument shall not overheat while operating 24 hours in an ambient temperature of 25°C, when restricted on all sides and the rear with a clearance of 2 inches. This may require an internal electromechanical device such as a fan or blower. If such a device is required, a through f shall apply.

a. The device shall be an integral, permanently attached part of the unit.

b. The device shall not require an additional power input source.

c. The air inlet shall be filtered against dust and the filter shall be easily accessible and removable for cleaning.

d. The device shall not interfere with any operation of the instrument through noise, electrical interference, or vibration.

e. A thermal switch or relay shall be provided which will interrupt line power to the unit when the internal temperature rises to the degree of possible damage to the instrument.

f. Commutated or brush type motors shall not be used.

3.7.4 Markings.

3.7.4.1 Panel. Panel markings exposed to operational wear shall be metal stamped, etched, or engraved and shall be filled with a permanent material which contrasts with the color of the instrument's panel.

3.7.4.2 Dials and switches. Dial and switch markings exposed to operational wear shall be metal stamped, etched, or engraved and shall be filled with a permanent material which contrasts with the color of the instrument's dials and switches.

3.7.4.3 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.7.5 Workmanship.

3.7.5.1 General. The instrument shall be capable of meeting the design and construction requirements specified in 3.2, and shall be free from any damage or defect that could result in failure to comply with the requirements of this specification.

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* 3.7.5.2 Detail. The workmanship shall be such that no evidence of any physical damage or defect shall be evident upon examination of assembly and fit, materials, parts, freedom from binding, wiring assembly (lacing) and insulation, soldering, impregnation of coils, markings and identification of parts, accessibility to interchangeable parts, freedom of parts from burrs and sharp edges, finishes, and external alignment of component parts and assemblies.

* 3.7.5.3 Controls. Operating and adjustment controls, both internal and external, shall be readily accessible, suitably arranged, and of such size and construction to permit convenience and ease of operation under all service conditions. Unless otherwise specified, operating controls shall be located on the front panel of the instrument. Selector switches shall have positive detent action and shall be aligned properly with their markings.

* 3.7.5.4 Connectors. Unless otherwise specified, all connectors for signal inputs and outputs shall be of standard commercial type and quality, preferably BNC, UHF, N, or banana type. Not more than 2 different type connectors shall be used throughout the instrument.

* 3.8 Environmental requirements. The instrument shall show no evidence of physical damage or circuitry failure and shall maintain the accuracies specified in 3.5, after being subjected to the following environmental conditions:

a. Temperature, operating - 0°C to plus 55°C.

b. Temperature, nonoperating - Minus 54°C to plus 75°C.

c. Vibration - Total displacement of 0.02 inches from 25 to 55 Hz and 0.04 inches from 10 to 25 Hz for 30 minutes at each of the instrument's resonant frequencies.

d. Shock - Dropping the instrument on a hardwood table top after each bottom horizontal edge of the top, bottom, and two sides is elevated 4 inches or when the horizontal axis forms an angle of 45 degrees with the table top, for a total of 16 drops.

e. Altitude, operating - 0 to 10,000 feet.

f. Altitude, nonoperating - 0 to 50,000 feet.

g. Humidity - 0 to 95 percent.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary, to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspection. Examination, sampling, and testing of the instruments shall be classified as follows:

a. Quality conformance inspection (4.4).

b. Qualification inspection (4.9).

* 4.3 Test conditions. Unless otherwise specified herein, the instruments shall be tested under the following conditions:

a. Temperature - Room ambient, 25°C, ±15°C.

b. Altitude - Normal ground.

c. Humidity - Room ambient up to 90 percent relative humidity.

d. Power Source - 115 volts rms, ±2 percent; 60 Hz, ±1 percent.

e. Attitude - Normal operating position.

* 4.4 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests (4.4.1).

b. Environmental tests (4.4.2).

* 4.4.1 Individual tests. After the instruments have been properly calibrated, each instrument shall be subjected to the individual tests specified in Table I, Group D. One instrument, selected at random from each production lot of 50 or less serially numbered instruments produced on a contract or order, shall be subjected to all the individual tests specified in Table I, Group A.

* 4.4.2 Environmental tests. One instrument, selected at random from each production lot of 50 or less serially numbered instruments produced on a contract or order, shall be subjected to all the environmental tests specified in Table I, Group B, Subgroup 1. The instrument selected for the environmental tests may be the same instrument which was subjected to the individual tests specified in 4.4.1.

* 4.4.3 Delivery and acceptance. The production lot from which the samples are selected shall not be submitted to the Government for acceptance and delivery until the samples have satisfactorily completed all the tests specified in Table I, Group A and Group B, Subgroup 1.

* 4.4.3.1 Rejection and retest. When 1 instrument from a production lot fails to meet the specification, instruments still on hand or later produced shall not be accepted until the cause of failure (see 3.2.4) is determined and corrected. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. An additional 3 samples shall be selected from the same lot and shall be subjected to that portion of the test which produced the failure.

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* 4.4.4 Continuation of quality conformance inspection. Quality conformance inspection may be continued pending the investigation of a sampling test failure. Final acceptance of instruments on hand or later produced shall not be made until the contractor determines the cause of failure and demonstrates to the satisfaction of the procuring agency that the instruments meet all the requirements of this specification.

TABLE I - Testing Index

Test		Requirement	Test Method
Group A:	Individual tests		
	a. Examination of product	3.7.5	4.6.1
	b. Warm up	3.6.1	4.6.1
	c. Power supply	3.5.1	4.6.2
	d. Accuracy and frequency response	3.5.2.2, 3.5.2.3	4.6.3.1
	e. AC meter tracking	3.5.2.4	4.6.3.2
	f. Input impedance	3.5.2.5	4.6.3.3
	g. Amplifier output and impedance	3.5.3.1	4.6.4.1
	h. Noise	3.5.3.3	4.6.4.2
	i. AC hum pickup	3.5.3.4	4.6.4.3
	j. Scale accuracy	3.5.4.2	4.6.5.1
	k. Switching transients	3.5.4.5	4.6.5.2
	l. Meter response	3.5.4.6	4.6.5.3
	m. Range selector switch	3.5.5	4.6.6
	n. Terminals	3.5.6	4.6.7
	o. Test lead(s)	3.5.7	4.6.8
Group B:	Environmental tests		
Subgroup 1:			
	a. Temperature, nonoperating	3.8	4.7.1
	b. Vibration and shock	3.8	4.7.2

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Subgroup 2:

a. Temperature, operating	3.8	4.7.3
b. Altitude, operating	3.8	4.7.3
c. Altitude, nonoperating	3.8	4.7.4
d. Humidity	3.8	4.7.3
Group C: Reliability tests		
Reliability	3.2.4	4.8
Group D: Individual tests		
a. Accuracy and frequency response	3.5.2.2, 3.5.2.3	4.6.3.1
b. AC meter tracking	3.5.3.3	4.6.3.2
c. Noise	3.5.3.3	4.6.4.2
d. AC hum pickup	3.5.3.4	4.6.4.3
e. Switching transients	3.5.4.5	4.6.5.2
f. Meter response	3.5.4.6	4.6.5.3

* 4.5 Equipment required. The following items of test equipment, or units of equal or greater accuracy and range, shall be used to perform the tests. The test equipment shall conform to laboratory standards whose calibration is traceable to the prime standards at the National Bureau of Standards (NBS).

Equipment	Model
Variable Frequency Power Supply	Empire Devices VP-1000
Voltmeter Calibrator	Hewlett-Packard E02-738BR
Battery Powered Oscillator	Hewlett-Packard 204C
Electronic Multimeter	Hewlett-Packard 410C
L-C Meter	Tektronix 130
Decade Resistor	General Radio 1432-N
Resistor, 10 megohm, ±1 percent	Any type capable of performing the required test
Stop watch	Any type capable of performing the required test

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Vibration Test Machine	Any type capable of performing the required test
Environmental Test Chamber	Any type capable of performing the required test

* 4.6 Test method, Group A. The following tests, or equivalent thereof (see 4.9.3), shall be performed. The tests need not be conducted in the sequence as specified herein.

4.6.1 Inspection and warm up.

* 4.6.1.1 Mechanical inspection. The instrument shall be visually and mechanically inspected to determine compliance with the requirements specified herein with respect to design and construction, workmanship, dimensions, weight, and markings.

* 4.6.1.2 Functional inspection. In the performance of all tests, the instrument shall be inspected to determine compliance with the control and functional requirements set forth in Sections 3 and 4 of this specification and to the specifications as stated by the manufacturer. Particular attention shall be paid to insure that the complete range of all control functions, within the tolerance, linearity, and stability requirements, can be utilized and will not necessarily be limited to the test points as specified in Section 4 of this specification.

* 4.6.1.3 Warm up. Connect the instrument to a variable frequency power supply. Adjust the power supply to obtain 115 volts, 60 Hz. Turn instrument power switch on and allow 2 minutes for warm up.

* 4.6.2 Power supply. There are no direct power supply tests. Satisfactory operation of the power supply shall be determined by other tests specified herein.

4.6.3 AC voltmeter section.

* 4.6.3.1 Accuracy and frequency response.

a. Using the frequency response test set, apply a 400 Hz signal to the instrument and adjust the amplitude for a reading of 0.9 on the 0.001 volt range.

b. Adjust the output frequency of the test set in turn, to 10 Hz, 20 Hz, 50 Hz, 1 MHz, 2 MHz, and 4 MHz. Maintain a constant amplitude with respect to the 400 Hz reference signal.

c. Note the meter reading at each frequency. The readings, referenced at 400 Hz, shall not exceed the tolerances specified in 3.5.2.3.

d. Repeat the above test on the following ranges: 0.003, 0.01, 0.03, 0.1, 0.3, 1, and 3, using the 0.9 mark on the 0 to 1 scale as a reference.

e. The accuracies shall be the same as in c.

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f. Select the voltage range in the above test which displays maximum error at 4 MHz.

g. Repeat steps a through c on this range, using input line voltages of 105 and 125 volts rms, 210 and 250 volts rms, and input line voltage frequencies of 50 and 400 Hz.

h. The combined error, due to power line and signal frequency variations, shall not exceed the tolerances specified in 3.5.2.3.

i. Return the line voltage and frequency to 115 volts rms, 60 Hz.

j. Using the frequency response test set, apply a 400 Hz signal to the instrument and adjust the amplitude for a reading of 0.9 on the 1 volt range.

k. Slowly sweep the frequency of the test set from 10 Hz to 4 MHz, maintaining a constant amplitude with respect to the 400 Hz reference signal.

l. Note the meter readings from 10 Hz to 4 MHz. The readings, referenced at 400 Hz, shall not exceed the tolerances specified in 3.5.2.3.

* 4.6.3.2 AC meter tracking.

a. Using a 400 Hz signal source, apply a full scale voltage to the instrument's 1 volt range.

b. Decrease the applied voltage in 0.1 volt steps from 1 volt to 0.1 volt and note the meter readings.

c. The accuracy of the readings shall be within ±1 percent of full scale.

* 4.6.3.3 Input impedance.

a. Using an AC signal source, apply a full scale voltage to the instrument's 1 volt range.

b. Connect a 10 megohm, ±1 percent resistor in series with the input voltage to the instrument.

c. With the resistor in series with the instrument's input, the meter indication shall be half scale or more.

d. Remove the applied voltage and resistor from the instrument's input.

e. Using an L-C meter, measure the capacitance across the input terminals of the instrument.

f. The capacitance shall be not more than 25 pf.

4.6.4 Amplifier.

* 4.6.4.1 Output voltage and impedance.

a. Using an AC signal source, apply a full scale voltage to the instrument's 1 volt range.

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b. Using an electronic multimeter, measure the open circuit voltage across the instrument's output terminals.

c. The open circuit voltage shall be at least 150 mv, rms.

d. Connect a decade resistor across the instrument's output terminals and in parallel with the electronic multimeter. Adjust the decade resistor until the multimeter indicates exactly 1/2 the voltage obtained in c.

e. The resistance value now indicated on the decade resistor is equal to the output impedance of the instrument, and shall be not greater than 60 ohms.

f. Apply a full scale voltage to the instrument's 0.1 volt range and measure the DC output.

g. The output shall be at least 1 volt DC.

h. Repeat step f on the 0.3, 1, and 3 volt ranges.

i. The output shall be at least 1 volt DC on each range.

4.6.4.2 Noise.

a. With the instrument's input terminals shorted together and the selector switch set to the 1 mv range, note the residual reading on the meter.

b. The residual reading shall be less than 3 percent of full scale.

c. Repeat the above test with the selector switch set to the 1 volt range.

4.6.4.3 AC hum pickup.

a. Using the battery operated audio oscillator, apply a 100 Hz signal to the instrument and adjust the amplitude for a reading of 0.9 on the 1 volt scale.

b. Slowly adjust the oscillator frequency to about 60 Hz and note the maximum deviation of the meter pointer.

c. Slowly adjust the oscillator frequency to about 120 Hz and note the maximum deviation of the meter pointer.

d. The maximum deviation of the meter pointer, due to beating with 60 Hz and 120 Hz, shall be less than ±1 percent of full scale.

4.6.5 Meter scales.

4.6.5.1 DB and voltage scale accuracy.

a. Using the frequency response test set, apply a 1000 Hz signal to the instrument's input and adjust the amplitude for a +2 dB reading on the instrument. Note the corresponding reading on the voltage scale.

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b. Vary the amplitude of the test set for meter readings of 0 dB, -2 dB, -4 dB, -6 dB, -8 dB, -10 dB, and -12 dB. For each dB increment, note the corresponding reading on the voltage scale.

c. Each dB increment shall correspond to the voltage scale values specified in 3.5.4.2, within ±1 percent of full scale.

d. The voltage scale accuracy shall be determined by other tests specified herein.

4.6.5.2 Switching transients.

a. With the instrument's input terminals shorted together, change the selector switch from the 0.3 volt range to the 1 volt range. Make every effort to prevent physical vibration of the instrument when changing selector switch positions.

b. The maximum deflection of the meter pointer, due to switching transients, shall be less than 25 percent of full scale.

c. Immediately following the change of the selector switch, the meter pointer shall return to zero reading in less than 10 seconds.

4.6.5.3 Meter response.

a. Using an AC signal source, apply a full scale voltage to the instrument's 1 volt range and measure the time required for the meter pointer to reach full scale deflection from zero.

b. The meter pointer shall reach full scale deflection in not more than 1.5 seconds.

4.6.6 Range selector switch. Visually and mechanically inspect the range selector switch for positions, markings, alignment, and action to assure compliance with the requirements specified in 3.5.5.

4.6.7 Terminals. Visually and mechanically inspect the input and output terminals to assure compliance with the requirements specified in 3.5.6.

4.6.8 Test lead(s). Satisfactory operation of the test lead(s) shall be determined by other tests specified herein.

4.7 Test method, Group B.

4.7.1 Temperature test, nonoperating.

4.7.1.1 Low temperature test. Place the instrument in a temperature test chamber and reduce the temperature of the chamber to minus 54°C. Maintain this temperature for 4 hours. Return the test chamber and instrument to room temperature in 60 minutes. Remove the instrument from the test chamber and immediately perform the test specified in 4.6.3.1. The instrument shall operate within the specified tolerances and there shall be no indication of failure as a result of the low temperature.

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* 4.7.1.2 High temperature test. Place the instrument in a temperature test chamber and elevate the temperature of the chamber to plus 75°C. Maintain this temperature for 4 hours. Return the test chamber and instrument to room temperature in 60 minutes. Remove the instrument from the test chamber and immediately perform the test specified in 4.6.3.1. The instrument shall operate within the specified tolerances and there shall be no indication of failure as a result of the high temperature.

* 4.7.2 Vibration and shock.

* 4.7.2.1 Vibration test. The instrument shall be rigidly attached, in the normal operation position, to a vibration testing mechanism. The resonant frequencies of the instrument shall be determined by varying the frequency of applied vibration slowly through the range specified in 3.8c at reduced amplitudes. Individual resonance searches shall be conducted with the vibration applied along each of the 3 mutually perpendicular axes of the instrument. The instrument shall then be subjected to a total displacement of 0.02 inches from 25 to 55 Hz and 0.04 inches from 10 to 25 Hz for 30 minutes at each of the instrument's resonant frequencies, and at 10 Hz and 55 Hz for 15 minutes at each frequency. At the completion of this test, perform the test specified in 4.6.3.1. The instrument shall operate within the specified tolerances and there shall be no evidence of physical damage or circuitry failure as a result of vibration.

* 4.7.2.2 Shock. The instrument shall be subjected to the following shock test on a hardwood table top. Using one edge of the instrument as a pivot, tilt the instrument until the horizontal axis of the assembly forms an angle of 45 degrees with the table top or until the elevated edge is 4 inches above the table top, whichever occurs first. Allow the instrument to drop freely to the horziontal. Repeat this procedure, using the other edges of the same horizontal face and the other 3 sides, excluding the front and rear, for a total of 16 drops. At the completion of this test, perform the test specified in 4.6.3.1. The instrument shall operate within the specified tolerances and there shall be no evidence of physical damage or circuitry failure as a result of this test.

* 4.7.3 Temperature, altitude, and humidity, operating.

* 4.7.3.1 Low temperature and altitude tests. Place the instrument is a test chamber and make the necessary connections for external measurements. Adjust the chamber conditions to 10,000 feet altitude and 0°C. Maintain this temperature and altitude for 2 hours. Apply power to the instrument and allow 2 minutes for warm up. Perform the test specified in 4.6.3.1. The instrument shall operate within the specified tolerances and there shall be no indication of failure as a result of this test. Return the test chamber and instrument to room temperature.

* 4.7.3.2 High temperature and humidity tests. Place the instrument in a test chamber and make the necessary connections for external measurements. Adjust the chamber conditions to 95 percent relative humidity and plus 55°C. Maintain this temperature and humidity for 20 hours. Apply power to the instrument and allow 2 minutes for warm up. Perform the test specified in 4.6.3.1. The instrument shall operate within the specified tolerances and there shall be no indication of failure as a result of this test. Return the test chamber and instrument to room temperature.

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* 4.7.4 Altitude, nonoperating. Place the instrument in a test chamber. Adjust the chamber conditions to 50,000 feet altitude. Maintain this altitude for 2 hours. Return the test chamber and instrument to normal ground and perform the test specified in 4.6.3.1. The instrument shall operate within the specified tolerances and there shall be no indication of failure as a result of this test.

* 4.7.5 At the completion of the above environmental tests, perform all the individual tests as specified in Table I, Group A. The instrument shall operate within the specified tolerances. There shall be no evidence of physical damage or circuitry failure as a result of these tests.

* 4.8 Test method, Group C. The following reliability tests shall be performed only on qualification test samples.

* 4.8.1 Accept-reject criteria. Acceptance and rejection for reliability qualification shall be based on the schedule given in Table II.

* 4.8.2 Input voltage cycling. The input voltage to the samples under test shall be varied in accordance with the following schedule: 125 volts for the first 24-hour cycle, 115 volts for the second 24-hour cycle, and 105 volts for the last 24-hour cycle. These 24-hour cycles shall be repeated until the accept-reject criteria in 4.8.1 are met. Continuous testing is not mandatory; these tests may be run in 8-hour intervals with the input voltages controlled to within ±2 percent. Concurrent testing of samples is encouraged.

TABLE II - Accept-Reject Criteria

No. of Failures	Total test time* Reject (Equal or less)	Total test time* Accept (Equal or more)
0	N/A	2.8
1	N/A	3.6
2	N/A	4.4
3	0.16	5.2
4	0.97	6.0
5	1.8	6.8
6	2.6	7.7
7	3.4	8.5
8	4.2	9.3
9	5.0	10.1
10	5.8	10.9
11	6.6	11.7
12	7.5	12.5
13	8.3	13.3
14	9.1	14.2
15	9.9	14.6
16	10.7	14.6
17	11.5	14.6
18	12.3	14.6
19	14.0	N/A

*Total test time is total unit hours of "equipment on" time and is expressed in multiples of the specified MTBF.

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* 4.8.3 Test intervals. The individual tests of Table I, Group D shall be run at 24-hour intervals of operation at the input voltage conditions as specified in 4.8.2. This series of tests shall be repeated as many times as necessary to reach a decision in accordance with 4.8.1. In the event that a sample failure is detected in the 24-hour interval test, and the exact time of failure cannot be ascertained, the sample shall be credited with 12 hours of successful operation for that particular 24-hour interval or 1/2 the total time of the interval. The operational mode to be used during operating periods, when the instrument is not undergoing specification checks, will be the test specified in 4.6.3.1. No calibration or adjustments shall be made once the test is started, beyond those allowed under individual test procedures. Operating time accumulated during the specification checks may be included as part of the total operating time.

* 4.9 Qualification inspection.

* 4.9.1 Qualification tests. Prior to actual contract award, all products submitted by any manufacturer shall have passed the qualification tests as specified in Table I, Group A, Group B, and Group C.

* 4.9.2 Qualification test samples. The qualification test samples shall consist of 4 instruments of each type, complete with operator's manual and schematic diagram. Samples shall be identified by the manufacturer's own type or model number. The manufacturer shall submit a complete alignment and calibration procedure with each sample. The tests, as sequenced in this specification, do not constitute proper calibration procedures. Qualification test samples shall have been subjected to and have passed the tests as specified in Table I, Group A, prior to submission of application for qualification testing. The manufacturer shall forward a test report, certified by a registered professional or chief engineer, to the Government with the application, including records of the tests for each unit submitted. These records shall show actual readings taken wherever applicable, rather than check marks indicating conformance.

* 4.9.3 Equivalent tests. If the manufacturer performs tests equivalent to those specified in 4.6, the test procedures shall be submitted to the Government for approval along with the application for qualification testing. The test procedures shall reference the equivalent test to the applicable requirement specified in 3.5. Further change or amendment to the test procedures by the manufacturer will require resubmittal to the Government for approval and acceptance.

* 4.9.4 Discontinuance of tests. The Government maintains the prerogative of stopping the qualification testing at any time a test sample fails to meet the specification requirements. Return of the test sample to the manufacturer may be required whenever the Government determines the sample has not complied with the specification requirements.

* 4.9.4.1 Adjustments during qualification testing. After the qualification tests have started, only adjustments to externally available controls, other than those specified in 3.2.3, will be permitted.

* 4.9.5 Resubmission. Samples should not be resubmitted for qualification testing within 6 months following return of the test samples to the manufacturer. A minimum of 6 samples may then be resubmitted for qualification testing upon receipt and evaluation by the Government of documented evidence of corrective action. A certified test report, in accordance with the

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specification requirements in 4.9.2, for all samples to be resubmitted for qualification testing, shall be delivered to the Government prior to resubmission of test articles. These test reports shall include all the tests performed and the test results.

4.9.6 Cost of tests. The Government will be responsible for conducting one qualification test cycle of a particular sample from a supplier at a Government operated or contract laboratory. The cost of additional tests shall be borne by the applicant at an independent laboratory designated by the Government.

4.10 Inspection of preservation, packaging, packing, and marking. The instruments shall be visually inspected to determine compliance with the requirements specified herein with respect to preservation, packaging, packing, and marking.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The equipment, accessories, and technical publications shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555, Level A, B, or C (see 6.2).

6. NOTES

6.1 Intended use. The instruments are intended for use in precision laboratory, field, shop, and depot maintenance applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type required.

c. Selection of applicable level of preservation, packaging, and packing.

6.3 Qualification. With respect to products requiring qualification, awards will made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the product covered by this specification. The activity responsible for the Qualified Products List is the Service Engineering Division, SANE, Kelly Air Force Base, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to teh last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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